

Pythagoras' Theorem & Trigonometry

Extra practice test 4 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $a^2 + b^2 = 10^2 + 24^2 = 100 + 576 = 676$ (1) $c = \sqrt{676} = \mathbf{26\text{ cm}}$ (1)

A2. $a^2 = 20^2 - 12^2$ (1) $= 400 - 144 = 256$ (1) $a = \sqrt{256} = \mathbf{16\text{ cm}}$ (1)

A3. $\text{diagonal}^2 = 18^2 + 24^2 = 324 + 576 = 900$ (1) $\text{diagonal} = \sqrt{900} = \mathbf{30\text{ cm}}$ (1)

Section B — SOH CAH TOA basics

B1. The **adjacent** side (1)

B2. **TOA**: $\text{opposite} = \text{adjacent} \times \tan \theta$ (1)

B3. **CAH**: $\theta = \cos^{-1}(\text{adjacent} \div \text{hypotenuse})$ (1)

Section C — Finding missing sides

C1. $\sin 33^\circ = x \div 16$ (1) $x = 16 \times \sin 33^\circ$ (1) $= \mathbf{8.7\text{ cm}}$ (1)

C2. $\cos 47^\circ = 6 \div h \rightarrow h = 6 \div \cos 47^\circ$ (1) $= \mathbf{8.8\text{ cm}}$ (1)

C3. $\tan 58^\circ = x \div 3 \rightarrow x = 3 \times \tan 58^\circ$ (1) $= \mathbf{4.8\text{ cm}}$ (1)

Section D — Finding missing angles

D1. $\sin \theta = 9 \div 15 = 0.6 \rightarrow \theta = \sin^{-1}(0.6)$ (1) $= \mathbf{36.9^\circ}$ (1)

D2. $\tan \theta = 13 \div 6 = 2.1667$ (1) $\theta = \tan^{-1}(2.1667)$ (1) $= \mathbf{65.2^\circ}$ (1)

Section E — Applying Pythagoras & trigonometry

E1. Use Pythagoras: $\text{height}^2 = 6.5^2 - 2.4^2 = 42.25 - 5.76 = 36.49$ (1) $\sqrt{36.49} = \mathbf{6.0\text{ m}}$ (1)

E2. $\tan 15^\circ = h \div 12$ (1) $h = 12 \times \tan 15^\circ$ (1) $= \mathbf{3.2\text{ m}}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.